

Product Decision Memo

Progression Gate Experiment: Level 30 vs Level 40

To: Product Manager

Subject: Impact of moving the first game gate from Level 30 to Level 40

Executive Summary

We tested whether moving the first progression gate from **Level 30 to Level 40** would improve player retention.

Around **90,000 players** were split between the existing Level 30 experience and the new Level 40 experience. The two groups were well balanced, so the comparison is reliable.

The key finding is that **moving the gate to Level 40 did not improve retention**. Day-1 retention was slightly lower, but the difference was not large enough to confidently attribute to the gate change. More importantly, **Day-7 retention was lower by 0.82 percentage points** for players experiencing the Level 40 gate.

On a scale of **1 million players**, this difference represents **approximately 8,183 fewer players returning on Day 7**.

What We Found

Metric	Level 30	Level 40	Difference
Day-1 retention	44.82%	44.23%	-0.59 pp
Day-7 retention	19.02%	18.20%	-0.82 pp

Day 1: The small difference could reasonably be normal player-to-player variation. We do not have enough evidence to say the gate change caused it.

Day 7: The difference is much more consistent and was unlikely to be explained by random variation alone. Our analysis gives strong evidence that the Level 40 experience had lower Day-7 retention.

We also repeated the analysis using **10,000 simulated samples**, and the same pattern appeared.

Data Quality

One player recorded **49,854 game rounds**, far outside the rest of the dataset. This was treated as an obvious data anomaly and removed before comparing retention.

After removing that one record, the two groups remained almost identical in size and gameplay behavior.

What This Means for the Product

The experiment does **not** show that moving the gate to Level 40 improves player retention.

Instead, the strongest signal we see is a potential **negative effect on Day-7 retention**.

However, retention is only one part of the product decision. We do not yet know whether the change affected:

- Revenue or player lifetime value
- In-game purchases
- Advertising revenue
- Longer-term retention
- Player engagement
- Different types of players differently

Recommendation

Do not make the Level 40 gate the default based on this experiment alone.

Before making a final decision, I would run additional analysis or testing covering:

1. **Day-14 and Day-30 retention**
2. **Revenue and lifetime value**
3. **Game engagement and session behavior**
4. **Player feedback / experience**
5. **Results across different player segments**
6. **A follow-up experiment to confirm whether the Day-7 effect persists**

Bottom Line

The Level 40 gate **did not deliver the expected retention improvement**. Day-1 retention was essentially inconclusive, while Day-7 retention was **0.82 percentage points lower** than the Level 30 experience.

At larger scale, that difference could represent thousands of fewer returning players. We should therefore investigate the longer-term and financial impact before adopting the Level 40 gate as the new standard.